NexSpace — Technical Stack & Architecture Guide

Complete Inventory of Frameworks, Machine Learning Models, Languages, and Infrastructure

DATE: SEPTEMBER 2026 | PLATFORM VERSION: v0.1.0 | REMOTE-SENSING VLM PLATFORM

1. Programming & Scripting Languages

Core development languages utilized across the frontend, machine learning backend, and infrastructure.

TypeScript (v5.9+)

LANGUAGE

Purpose: Powers full-stack Next.js web application, React UI components, Server API proxy routes, and database abstraction services.

Importance: CRITICAL: Enforces strict end-to-end type safety, guarantees API contract parity with the Python ML microservice, and prevents runtime exceptions.

Python (v3.11.x)

LANGUAGE

Purpose: Powers the entire machine learning inference engine, neural vision backbones, FastAPI microservice, and geospatial math routines.

Importance: CRITICAL: The industry-standard runtime for deep learning frameworks (PyTorch, Transformers), raster manipulation, and scientific computing.

JavaScript / Node.js (v20+)

LANGUAGE

Purpose: Executes automation scripts, PDF audit generators, visual proof capture engines, and concurrent task runners.

Importance: HIGH: Automates development lifecycles, runs end-to-end verification suites, and produces exportable client deliverables.

SQL & PostGIS Spatial DDL

LANGUAGE / QUERY

Purpose: Defines relational database models and executes spatial polygon and bounding-box queries for satellite scenes.

Importance: HIGH: Enables geographical indexing, scene filtering, entity bounding-box storage, and geospatial audit logging.

HTML5, CSS3 & PostCSS

LANGUAGE / STYLING

Purpose: Constructs semantic document structure and cybernetic terminal visual design using modern dark-mode palettes.

Importance: HIGH: Delivers a responsive, accessible, and high-performance user interface for satellite intelligence inspection.

2. Full-Stack Web & UI Frameworks

Modern frontend architecture, UI components, and rendering engines.

Next.js 16 (App Router)

FRONTEND FRAMEWORK

Purpose: Acts as the primary application server, providing server-side rendering, API route proxies, dynamic routing, and security headers.

Importance: CRITICAL: Serves user-facing terminals, coordinates authentication, and proxies heavy ML requests to the backend.

React 19

UI COMPONENT LIBRARY

Purpose: Manages stateful UI components, high-frequency telemetry streams, live bounding-box overlays, and image viewer canvases.

Importance: CRITICAL: Powers dynamic and reactive geospatial analysis interfaces with concurrent rendering support.

Tailwind CSS v4

STYLING FRAMEWORK

Purpose: Provides utility classes for glassmorphism, responsive grid layouts, color palettes, and typography tokens.

Importance: HIGH: Ensures a consistent, highly performant dark-mode theme across all pages without custom CSS bloat.

Framer Motion

ANIMATION LIBRARY

Purpose: Drives fluid user interaction animations, panel sliding, evidence drawers, and transition states.

Importance: MEDIUM-HIGH: Enhances analyst experience through tactile micro-interactions and smooth UI feedback.

OGL (WebGL Minimal Engine)

GRAPHICS / CANVAS

Purpose: Renders GPU-accelerated particle shaders and orbital energy background visualizations on the dashboard.

Importance: MEDIUM: Delivers high-framerate visual enhancements directly on the GPU without degrading CPU thread performance.

3. Machine Learning & Geospatial Processing Frameworks

High-performance deep learning computation, server microservices, and geospatial analysis.

FastAPI

BACKEND FRAMEWORK

Purpose: High-performance asynchronous Python REST API server hosting all ML inference routes on port 8000.

Importance: CRITICAL: Provides low-latency, non-blocking execution endpoints for heavy vision-language model requests.

PyTorch (torch >= 2.1)

DEEP LEARNING RUNTIME

Purpose: Underlying tensor computation and hardware acceleration engine running model weights on CUDA GPU and multi-core CPU.

Importance: CRITICAL: Executes forward passes for vision transformers, object grounding, and feature extraction.

Hugging Face Transformers

MODEL ARCHITECTURE SUITE

Purpose: Loads pretrained model weights, processors, tokenizers, and conditional generation pipelines.

Importance: CRITICAL: Standardizes multi-modal model loading, inference configurations, and processor pipelines.

NumPy & SciPy

SCIENTIFIC COMPUTING

Purpose: Performs matrix operations, Otsu thresholding, spatial gradients, Pearson correlations, and vegetation indices (NDVI/ExG).

Importance: CRITICAL: Powers mathematical change detection algorithms and multi-sensor alignment calculations.

Pillow (PIL) & tifffile

RASTER PROCESSING

Purpose: Decodes multi-spectral imagery, GeoTIFF satellite files, and coordinates bounding-box image cropping.

Importance: HIGH: Ingests raw overhead satellite files and prepares them for tensor pipelines.

Prisma ORM

DATABASE ORM

Purpose: Type-safe database client and schema migration manager for PostgreSQL.

Importance: HIGH: Ensures secure and consistent database read/write queries for user accounts and scene registries.

4. Artificial Intelligence & Vision-Language Models

Specialized remote-sensing, zero-shot detection, and multi-sensor fusion models.

flax-community/clip-rsicd

VISION-LANGUAGE MODEL

Provenance: REAL RS-ADAPTED MODEL (Trained on RSICD remote-sensing benchmark)

Purpose: Zero-shot remote-sensing scene and land-cover classification over standardized taxonomies (ports, runways, urban, cropland).

Importance: CRITICAL: Overcomes nadir perspective domain shifts to accurately classify aerial scenes without generic photo bias.

IDEA-Research/grounding-dino-tiny

OBJECT GROUNDING MODEL

Provenance: REAL ZERO-SHOT OBJECT DETECTION MODEL

Purpose: Open-vocabulary object localization detecting arbitrary text prompts (vessels, oil tanks, aircraft) into normalized [0, 1000] boxes.

Importance: CRITICAL: Identifies and localizes critical geospatial infrastructure and tactical assets directly from natural language.

google/paligemma-3b-ft-rsvqa-lr-224

REMOTE-SENSING VQA

Provenance: REAL RS-ADAPTED VLM (Fine-tuned on RSVQA dataset)

Purpose: Visual question answering engine answering domain-specific inquiries regarding object count, roads, waterways, and terrain.

Importance: CRITICAL: Enables interactive dialogue and visual reasoning over satellite scenes.

Salesforce/blip-image-captioning-base

VISION TRANSFORMER BACKBONE

Provenance: GENERIC PRETRAINED MODEL (ViT-B / Text Decoder)

Purpose: Generates natural language optical captions and extracts 768-dimensional vision embeddings for multi-sensor fusion.

Importance: HIGH: Provides textual summaries and acts as the shared optical feature encoder for Optical + SAR fusion.

Claude 3.5 Sonnet (Anthropic API)

FRONTIER LLM

Provenance: FRONTIER MULTI-MODAL REASONING API

Purpose: Parses complex natural language user queries into structured geospatial filters (location AOI, sensors, date range, entity types).

Importance: HIGH: Translates unstructured human language into structured database parameters and task execution strategies.

5. Specialized Geospatial Engines & Fusion Algorithms

Algorithmic pipelines, differential change analyzers, and radar-optical fusion.

Optical + SAR Physical Fusion Engine

MULTIMODAL FUSION ENGINE

Provenance: RESEARCH BASELINE (Cross-Modal ViT + Microwave Backscatter)

Purpose: Synthesizes optical RGB with microwave SAR backscatter (speckle CV, dynamic range, spatial correlation) to reveal hidden features.

Importance: HIGH: Combines optical visual clarity with radar all-weather cloud and foliage penetration.

Bi-Temporal Semantic Change Engine

CHANGE REASONING ENGINE

Provenance: RESEARCH BASELINE (Multi-Temporal Otsu & Grounding)

Purpose: Performs differential matrix comparison between two temporal passes to pinpoint construction, destruction, or vegetation loss.

Importance: HIGH: Enables automated damage assessment, environmental monitoring, and urban expansion tracking.

Sliding-Window High-Res Tiling & NMS Engine

GEOSPATIAL TILING ENGINE

Provenance: REAL GEOSPATIAL ENGINE

Purpose: Slices gigapixel satellite rasters into overlapping tiles, runs batch model inference, and unifies boxes with Non-Maximum Suppression.

Importance: HIGH: Allows small object detection (e.g. vehicles, small boats) across large multi-gigabyte satellite scenes.

6. Infrastructure & Container Services (Docker Compose)

Storage, database, and caching infrastructure.

PostgreSQL 16 + PostGIS 3.4 (nexspace_postgres)

INFRASTRUCTURE

Purpose: Relational database storing user records, scene metadata, detected geospatial entities, and coordinate polygon geometries.

Importance: CRITICAL: Provides robust persistent storage and spatial indexing for geospatial intelligence queries.

Redis 7 Alpine (nexspace_redis)

INFRASTRUCTURE

Purpose: High-speed in-memory caching for model inference states, telemetry counters, and asynchronous job queues.

Importance: HIGH: Delivers sub-millisecond response caching for repetitive user queries.

MinIO S3 Object Store (nexspace_minio)

INFRASTRUCTURE

Purpose: Self-hosted S3-compatible object store managing multi-gigabyte satellite GeoTIFF rasters and generated heatmaps.

Importance: HIGH: Provides scalable, high-throughput storage for overhead imagery and evidence artifacts.